

Calculus 1 Survival Sheet

Definition of the Derivative

The derivative of a function $f(x)$ at a point x is defined as:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

This limit, if it exists, gives the slope of the tangent line to the curve at the point x .

Basic Derivatives

$$\frac{d}{dx}(c) = 0 \quad (\text{constant rule})$$

$$\frac{d}{dx}(x) = 1$$

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

$$\frac{d}{dx}(e^x) = e^x$$

$$\frac{d}{dx}(\ln x) = \frac{1}{x}$$

$$\frac{d}{dx}(\sin x) = \cos x$$

$$\frac{d}{dx}(\cos x) = -\sin x$$

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

$$\frac{d}{dx}(\sec x) = \sec x \tan x$$

$$\frac{d}{dx}(\csc x) = -\csc x \cot x$$

$$\frac{d}{dx}(\cot x) = -\csc^2 x$$

Constant Multiple Rule

$$\frac{d}{dx}[c \cdot f(x)] = c \cdot f'(x)$$

Rules of Differentiation

Sum and Difference Rule

$$\frac{d}{dx}[f(x) \pm g(x)] = f'(x) \pm g'(x)$$

Product Rule

$$\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x)$$

Quotient Rule

$$\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$$

Chain Rule

If $y = f(g(x))$, then:

$$\frac{dy}{dx} = f'(g(x)) \cdot g'(x)$$

Example: If $y = \sin(x^2)$, then:

$$\frac{dy}{dx} = \cos(x^2) \cdot 2x$$

Implicit Differentiation

When y is defined implicitly as a function of x , differentiate both sides with respect to x , applying the chain rule to y terms.

Example: For the equation $x^2 + y^2 = 25$:

$$\begin{aligned} \frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) &= \frac{d}{dx}(25) \\ 2x + 2y \frac{dy}{dx} &= 0 \\ \frac{dy}{dx} &= -\frac{x}{y} \end{aligned}$$

Useful Derivatives

- $\frac{d}{dx}(a^x) = a^x \ln a$
- $\frac{d}{dx}(\arcsin x) = \frac{1}{\sqrt{1-x^2}}$
- $\frac{d}{dx}(\arccos x) = \frac{-1}{\sqrt{1-x^2}}$
- $\frac{d}{dx}(\arctan x) = \frac{1}{1+x^2}$

More Trig Derivatives

$$\frac{d}{dx}(\operatorname{arccot} x) = -\frac{1}{1+x^2}, \quad \frac{d}{dx}(\operatorname{arcsec} x) = \frac{1}{|x|\sqrt{x^2-1}}$$
$$\frac{d}{dx}(\operatorname{arccsc} x) = -\frac{1}{|x|\sqrt{x^2-1}}$$

Basic Integration Rules

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$$
$$\int \frac{1}{x} dx = \ln |x| + C$$
$$\int e^x dx = e^x + C$$
$$\int a^x dx = \frac{a^x}{\ln a} + C$$

Trig Integrals

$$\int \sin x dx = -\cos x + C$$
$$\int \cos x dx = \sin x + C$$
$$\int \sec^2 x dx = \tan x + C$$
$$\int \csc^2 x dx = -\cot x + C$$
$$\int \sec x \tan x dx = \sec x + C$$
$$\int \csc x \cot x dx = -\csc x + C$$

Integration by u -Substitution

If $u = g(x)$, then

$$\int f(g(x))g'(x) dx = \int f(u) du.$$

Steps:

1. Choose u (inside part).
2. Compute du .

3. Rewrite integral in u .
4. Integrate.
5. Substitute back.

Definition of $\ln x$ and e^x

$$\ln x = \int_1^x \frac{1}{t} dt.$$

The function e^x is the inverse of $\ln x$.

Summation Rules

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}$$

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\sum_{i=1}^n i^3 = \left(\frac{n(n+1)}{2} \right)^2$$

$$\sum_{i=1}^n (af(i) + bg(i)) = a \sum_{i=1}^n f(i) + b \sum_{i=1}^n g(i)$$

$$\sum_{i=1}^n c = cn$$

$$\sum_{i=k}^n f(i) = \sum_{j=1}^{n-k+1} f(j+k-1)$$

Riemann Sums

$$\text{Left sum: } L_n = \sum_{i=1}^n f(x_{i-1}) \Delta x$$

$$\text{Right sum: } R_n = \sum_{i=1}^n f(x_i) \Delta x$$

$$\text{Midpoint: } M_n = \sum_{i=1}^n f\left(\frac{x_{i-1} + x_i}{2}\right) \Delta x$$

$$\Delta x = \frac{b-a}{n}, \quad x_i = a + i\Delta x$$

Linearization / Tangent Line Approximation

$$L(x) = f(a) + f'(a)(x - a)$$

Approximation:

$$f(x) \approx L(x)$$

Fundamental Theorem of Calculus

Part 1

If $F'(x) = f(x)$, then

$$\frac{d}{dx} \int_a^x f(t) dt = f(x).$$

Part 2

$$\int_a^b f(x) dx = F(b) - F(a).$$

Continuity

A function is continuous at $x = a$ if:

$$\lim_{x \rightarrow a} f(x) = f(a).$$

Closed Interval and Extreme Values

If f is continuous on $[a, b]$, then it has an absolute max and an absolute min on $[a, b]$.

Concavity and Inflection Points

$$f''(x) > 0 \Rightarrow \text{concave up}, \quad f''(x) < 0 \Rightarrow \text{concave down}.$$

Inflection point: where concavity changes and $f''(x) = 0$ or undefined.

Mean Value Theorem

If f is continuous on $[a, b]$ and differentiable on (a, b) , then

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Rolle's Theorem

If $f(a) = f(b)$ and f satisfies MVT conditions, then

$$f'(c) = 0$$

for some c in (a, b) .

Tangent Line Formula

Slope: $m = f'(a)$ Equation:

$$y - f(a) = f'(a)(x - a)$$

Reversing Limits of Integration

If the bounds of an integral are reversed:

$$\int_b^a f(x) dx = - \int_a^b f(x) dx.$$

This is super helpful when simplifying definite integrals or matching the Fundamental Theorem of Calculus.

Even and Odd Functions in Integration

Even Functions

A function is **even** if:

$$f(-x) = f(x).$$

Examples: $\cos x$, x^2 , $|x|$

Integral shortcut:

$$\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx.$$

Odd Functions

A function is **odd** if:

$$f(-x) = -f(x).$$

Examples: $\sin x$, x^3 , $\tan x$

Integral shortcut:

$$\int_{-a}^a f(x) dx = 0.$$

Integrals from 0 to x and Variable Upper Limits

If

$$F(x) = \int_0^x f(t) dt,$$

then by the Fundamental Theorem of Calculus:

$$F'(x) = f(x).$$

More generally:

$$\frac{d}{dx} \left(\int_a^{g(x)} f(t) dt \right) = f(g(x)) \cdot g'(x).$$

And if the lower limit varies:

$$\frac{d}{dx} \left(\int_{g(x)}^a f(t) dt \right) = -f(g(x)) \cdot g'(x).$$

Integrals From $-a$ to a

Use symmetry when possible:

$$\int_{-a}^a f(x) dx = \begin{cases} 0, & f(x) \text{ odd} \\ 2 \int_0^a f(x) dx, & f(x) \text{ even} \end{cases}$$

This is especially useful in trig integrals and absolute value integrals.

Related Rates Templates

General Steps

1. Draw a diagram.
2. Identify all variables.
3. Write an equation relating quantities.
4. Differentiate both sides w.r.t. time t .
5. Substitute known values.
6. Solve for the desired rate.

Cone

Volume:

$$V = \frac{1}{3} \pi r^2 h.$$

If radius and height are proportional, use similar triangles before differentiating.

Similar Triangles

If two triangles are similar:

$$\frac{x}{a} = \frac{y}{b}.$$

Differentiate after expressing one variable in terms of the other.

Pythagorean (Ladder)

$$x^2 + y^2 = L^2.$$

Differentiate:

$$2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0.$$

Area Changing

$$A = \frac{1}{2}bh, \quad A = \pi r^2, \quad A = s^2.$$

Intermediate Value Theorem (IVT)

If f is continuous on $[a, b]$ and N is between $f(a)$ and $f(b)$, then there exists some $c \in (a, b)$ such that

$$f(c) = N.$$

Use IVT to show existence only (not to find c).

Steps:

1. Verify continuity on $[a, b]$
2. Compute $f(a)$ and $f(b)$
3. Show the target value lies between them

Inverse Trig Integrals

$$\begin{aligned}\int \frac{1}{a^2 + x^2} dx &= \frac{1}{a} \arctan\left(\frac{x}{a}\right) + C \\ \int \frac{1}{\sqrt{a^2 - x^2}} dx &= \arcsin\left(\frac{x}{a}\right) + C \\ \int \frac{1}{|x|\sqrt{x^2 - a^2}} dx &= \left(\frac{|x|}{a}\right) + C\end{aligned}$$

Tip: Match the form exactly before integrating.

Exponential Growth and Decay

$$\frac{dP}{dt} = kP$$

Solution:

$$P(t) = P_0 e^{kt}$$

- $k > 0$ growth
- $k < 0$ decay

Doubling time:

$$T = \frac{\ln 2}{k}$$

Half-life:

$$T = \frac{\ln 2}{|k|}$$

Differential Equations

Separable Equations

$$\frac{dy}{dx} = g(x)h(y)$$

Steps:

1. Separate variables
2. Integrate both sides
3. Add C
4. Solve for y

Initial Value Problems

Use the given condition (ex: $y(0) = y_0$) to solve for C .

Important: Plug in the initial value *after* integrating.

Trig Substitution

Expression	Substitution
$\sqrt{a^2 - x^2}$	$x = a \sin \theta$
$\sqrt{a^2 + x^2}$	$x = a \tan \theta$
$\sqrt{x^2 - a^2}$	$x = a \sec \theta$

Steps:

1. Substitute
2. Use trig identities
3. Integrate
4. Convert back using a triangle

Optimization Problems

Common goals: maximum area, minimum cost, shortest distance.

Steps:

1. Draw a diagram
2. Write the objective function
3. Use constraints to reduce to one variable
4. Find critical points
5. Check endpoints / reasonableness

Common setups:

- Fencing problems
- Box problems
- Distance problems